

Schedule/course outline

Week 1 Introduction to Computer Vision, motivation and basics

Week 2 Image Filtering

Week 3 Edge Detection

Week 4 Feature extraction methods

Week 5 Neural Networks

Week 6 CNN-based Computer Vision

Week 7 Generative Models

Week 8 Deep models in computer vision

Week 9 Sequence Modeling

Week 10 Representation Learning

Week 11 Object Detection and Image Segmentation

Week 12 Motion Estimation, Optical Flow

Week 13 Multi-model learning

Week 14 Visual-Language Models

Marks Distribution

50% final exam

20% Assignments

15% Midterm

10% Project

5% Quizzes

Late Policy

0 for late submission

Textbook

There is no text book for this class.

Suggested reference books are

- Mubarak Shah, “[Fundamentals of Computer Vision](#)“
- Ian Goodfellow, Yoshua Bengio and Aaron Courville, “[Deep Learning](#)“
- Richard Szeliski, [Computer Vision: Algorithms and Applications](#), Springer, 2010
(online draft),

Leading Journals and Conferences in Computer Vision

- [IEEE Computer Society Conference on Computer Vision and Pattern Recognition \(CVPR\)](#)
- [International Conference on Computer Vision \(ICCV\)](#)
- [European Conference on Computer Vision \(ECCV\)](#)
- [Pattern Analysis and Machine Intelligence \(PAMI\)](#)
- [International Journal of Computer Vision \(IJCV\)](#)